

Cycle120 ABF-Facing RS-MCA Counterexample Note

Date: 2026-06-24

Abstract

Let

```
K = F_17^32,  
H = <theta> subset K^*,  
|H| = 512,  
C = RS[K, H, 256].
```

Assuming the imported Cycle84 finite census and the Cycle116 fixed-jet transfer certificate, the row satisfies

```
LD_sw(C, 262) >= 52,747,567,092.
```

The printed ABF grand MCA definition uses the closed support threshold

```
|S| >= (1-delta)n.
```

At $n=512$ and $\delta=125/256$, this threshold is exactly 262. Therefore, under ABF Definition 4.3,

```
epsilon_mca(C, 125/256)  
>= 52,747,567,092 / 17^32  
> 2^-128.
```

Thus the certificate is an ABF-admissible negative certificate at

$\delta=125/256$: for this row, any largest value δ^*_C satisfying

$\epsilon_{\text{mca}}(C, \delta^*_C) \leq 2^{-128}$ must be strictly below $125/256$, provided the finite proof chain is correct.

Cycle119 strengthens the same row to agreement 263, hence distance at most $249 < 250 = (125/256)512$. This strict-ball strengthening is not needed for the printed ABF closed-threshold definition, but it is useful against stricter external conventions.

This note does not claim the exact value of δ^*_C , ordinary list decoding, protocol soundness failure, an asymptotic theorem, or peer-reviewed prize acceptance.

1. ABF Source Gates

Source:

```
Open Problems in List Decoding and Correlated Agreement  
Gal Arnon, Dan Boneh, Giacomo Fenzi  
IACR ePrint 2026/680  
local PDF SHA-256:  
e543ec6a4f3312b4383000e72e5aa23862e79cc9770ce21db2c48db679581de3
```

The ABF grand MCA challenge asks for a Reed-Solomon code

$C = \text{RS}[F, L, k]$ over a smooth evaluation domain $L \subset F$, with rate in $\{1/2, 1/4, 1/8, 1/16\}$, and threshold $\epsilon^* = 2^{-128}$.

The row

```
C = RS[F_17^32, H, 256],  
|H| = 512
```

passes the printed gates:

```
Field: finite field K = F_17^32.  
Domain: H = <theta> is a multiplicative subgroup of K^*.  
Smoothness: |H| = 512 = 2^9, so H is a power-of-two subgroup.  
Rate: 256/512 = 1/2.  
Envelope: 256 <= 2^40 and 17^32 < 2^256.
```

ABF's notation section states that $x \leftarrow S$ means uniform sampling from S .

Definition 4.3 samples

```
gamma <- F.
```

For this row, the MCA line parameter is therefore sampled uniformly from

$K = F_{17}^{32}$. No independent q_{chal} appears in the ABF grand MCA definition.

2. ABF MCA Predicate

ABF Definition 4.3 defines $\epsilon_{\text{mca}}(C, \delta)$ as the maximum over words

f_1, f_2 of the probability over $\gamma \leftarrow F$ that there exists a support S

with:

```
|S| >= (1-δ)n,  
Δ_S(f1 + γ f2, C) = 0,  
Δ_S((f1, f2), C^{==2}) > 0.
```

Equivalently, on the same support S :

```
f1 + γ f2 is exactly explained by a codeword of C,  
but f1 and f2 are not simultaneously explained by codewords of C.
```

This is exactly the support-wise same-support noncontainment predicate used in the Cycle116 and Cycle119 finite certificates.

3. Imported Finite Theorem

The Cycle84 finite census supplies

```
N = 52,747,567,092
```

distinct product values on the designated Cycle84 shell. The banked finite

receipt includes:

```
maximum product fiber size = 2,  
exactly 12 double fibers,  
no product fiber of size >= 3,
```

all 336 slot bridge identities.

The Cycle116 fixed-jet transfer proves a finite support-wise theorem for the smooth row

$$C = \text{RS}[F_{17^{32}}, H, 256], \quad |H|=512:$$
$$\text{LD}_{\text{sw}}(C, 262) \geq N.$$

Unpacked into ABF Definition 4.3, this means there are words f_1, f_2 in K^H and at least N distinct values γ in K such that for each such γ there is a support $S_\gamma \subseteq H$ satisfying:

$$|S_\gamma| \geq 262,$$
$$(f_1 + \gamma f_2)|_{S_\gamma} \text{ is code-explained by } C,$$
$$(f_1, f_2)|_{S_\gamma} \text{ is not simultaneously code-explained by } C^{\{=2\}}.$$

Thus the finite certificate gives at least N ABF-bad line parameters.

4. Threshold Arithmetic

For $n=512$ and $\delta=125/256$,

$$(1-\delta)n$$
$$= (1 - 125/256) * 512$$
$$= (131/256) * 512$$
$$= 262.$$

Therefore the Cycle116 agreement-262 theorem meets ABF Definition 4.3 exactly.

The field size is

$$17^{32} =$$
$$2367911594760467245844106297320951247361.$$

The exact threshold comparison is:

$$\text{floor}(17^{32} / 2^{128}) = 6,$$
$$52,747,567,092 > 6.$$

Equivalently,

$$2^{128} * 52,747,567,092 > 17^{32},$$

so

$$52,747,567,092 / 17^{32} > 2^{-128}.$$

Since ABF samples γ uniformly from K , the ABF MCA probability is at least this quotient.

5. Consequence For The Grand MCA Challenge

ABF says that resolving the grand MCA challenge for a code requires specifying the largest $\delta * C$ such that

$$\epsilon_{\text{mca}}(C, \delta * C) \leq 2^{-128}$$

and proving the corresponding upper and lower sides. The present packet proves only the negative side at one concrete radius:

$$\text{epsilon_mca}(C, 125/256) > 2^{-128}.$$

Therefore, for this row,

$$\text{delta_C}(2^{-128}) < 125/256$$

provided the finite proof chain is correct. This is prize-facing partial progress and a candidate counterexample to any expectation that this row remains MCA-secure through radius 125/256.

6. Cycle119 Strict Strengthening

Cycle119 proves a stronger finite/source theorem on the same row:

$$\text{LD_sw}(C, 263) \geq 52,747,567,092.$$

It uses a two-ended locator theorem rather than the invalid naive multiplication padding argument. The two-ended evaluator uses:

common top six locator coefficients,
one common nonzero constant coefficient,
selected product degrees 0, 250, 251, 252, 253, 254, 255.

It does not assume a common seventh top coefficient.

The resulting witnesses have distance at most

$$512 - 263 = 249,$$

which is strictly below

$$(125/256) * 512 = 250.$$

Thus Cycle119 is a one-symbol strict-ball strengthening. It is not needed for the ABF printed closed condition, but it should remain in the packet as a robustness addendum.

7. Proof Spine To Audit

The shortest proof spine is:

```
Cycle84 exact product occupancy
-> fixed-jet/product-scalar bridge
-> Cycle116 agreement-262 smooth RS transfer
-> ABF Definition 4.3 at delta=125/256
-> N / 17^32 > 2^-128.
```

The optional strengthening is:

```
Cycle84 exact product occupancy
-> two-ended Cycle119 transfer
-> agreement-263 strict-ball witnesses
-> same ABF density lower bound.
```

8. Non-Claims

This packet does not prove:

- the exact value of δ^*_C ;
- ordinary fixed-word list decoding;
- the ABF grand list-decoding challenge;
- protocol soundness failure;
- an asymptotic theorem;
- a deployed prime-field theorem;
- peer-reviewed acceptance;
- automatic prize award;
- an independent q_{chal} theorem for a separate protocol experiment.

9. Submission Implications

The Proximity Prize website states that a submission should include a PDF that:

- clearly states the claimed results,
- explains how they relate to the prize challenge,
- situates them with respect to prior work.

It also states:

- AI-aided submissions are allowed,
- human authors are responsible for correctness,
- submissions should be public on arXiv or IACR ePrint for timestamping,
- peer-reviewed acceptance is required for final prize consideration,
- partial progress may be considered,
- prize money may be split among multiple submissions.

Accordingly, this packet should be turned into a short human-edited PDF note before any public timestamp or prize email.